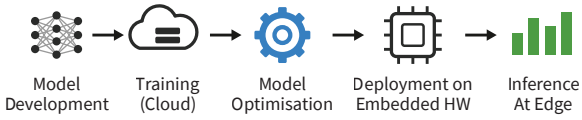
AI/ML WORKFLOW OVERVIEW



- ✓ Understand model types: CNN, RNN, Transformer, SVM etc.
- ✓ Focus on Inference, not Training, on embedded devices
- ✓ Model optimisation: Quantisation, Pruning, Knowledge Distillation
- ✓ Use frameworks: TensorFlow Lite, ONNX, PyTorch Mobile

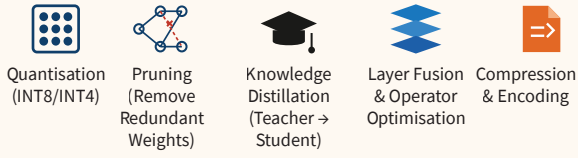
KEY HARDWARE BUILDING BLOCKS

- **CPU** — General purpose control, OS, application management
- **NPU / AI Accelerator** — High performance, low power inference
- **DSP** — Efficient for signal processing and neural networks
- **GPU** — Parallel processing for complex workloads
- **MCU** — Real-time control, low power management
- **Memory** — SRAM (on-chip), DRAM (LPDDR), Flash (NAND / NOR)
- **I/O Interfaces** — CSI/DSI, PCIe, USB, Ethernet, CAN, SPI, I2C, UART, GPIO

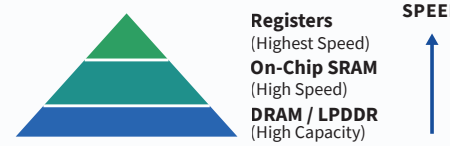
IMPORTANT DESIGN CONSIDERATIONS

- **Performance** — Meet real-time inference requirements (FPS, Latency)
- **Power Efficiency** — Optimise for mW / TOPS (Thermal and Battery life)
- **Memory Bandwidth** — Ensure sufficient bandwidth for data and model access
- **Thermal Management** — Heat dissipation for stable AI performance
- **Scalability** — Support future models and feature upgrades
- **Cost & BOM Optimisation** — Balance performance and cost
- **Security** — Secure Boot, Encryption, IP Protection
- **Reliability** — EMI/EMC, ESD, Surge, Functional Safety

MODEL OPTIMISATION TECHNIQUES



MEMORY HIERARCHY (TYPICAL)



TOOLS AND ECOSYSTEM

TensorFlow Lite	PyTorch Mobile	ONNX	Qualcomm AI Engine
arm CMSIS-NN	NVIDIA TensorRT	Cadence	altium MATLAB & Simulink

KEY TAKEAWAY

Great AI/ML experiences come from the right combination of algorithms, models, and hardware.

will always be connected. Hardware designers must understand both wired and wireless communication interfaces inside out. Crucially, you must know how integrating these interfaces impacts the surrounding hardware and overall system stability.

Once an edge device is deployed and connected, the ongoing cost of communication becomes a critical operational factor. This financial overhead cannot be an afterthought. It must be factored directly into the hardware design goals from day one.

Security has officially shifted from a theoretical, academic topic to a basic commercial necessity. AI edge devices are highly vulnerable targets, and designers must understand how to implement secure architectures at the hardware level to protect sensitive data and models.

The barrier between hardware and software has dissolved. Hardware engineers must know software now. You need a solid understanding of real-time operating systems and embedded Linux, specifically learning how to tune the operating system to maximise hardware performance.

Without a firm grasp of linear algebra and statistics, a hardware designer can never accurately size a controller or properly evaluate the mysterious neural processing unit (NPU).

The bottom line

This paradigm shift represents a major wake-up call, particularly for engineers who have overlooked the fundamental importance of mathematics and core electronics physics. While the transition to edge AI brings numerous complex challenges, mastering these seven areas forms the absolute baseline requirement for any hardware designer looking to stay relevant in the industry. EFY